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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : APM-301

BASIC MECHANICAL PROCESSING OF TEXTILES

Time Allotted : 3 Hours | *Full Marks : 70*

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP – A

(Answer all questions)

1. A. Choose the correct alternatives for the following : $5 \times 1 = 5$

i) The length of 2 kg 20 Tex cotton yarn is

a) 90 km

b) 100 km

c) 180 km

d) 360 km

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iii) The maximum number of interlacement can be obtained in

a) Plain weave b) 2/1 twill weave

c) 3/1 twill weave d) 5 end sateen weave.

iii) The direct yarn count of 35 Ne cotton yarn is

a) 140 Denier

b) 150 Denier

c) 180 Denier

d) 315 Denier.

iv) Ramie is a

a) Leaf fibre

b) Bast fibre

c) Seed hair

d) Synthetic fibre.

v) Shuttle is related to

a) Shedding

b) Picking

c) Beat up

d) Take up.

B. Match the **left hand side** with **right hand side** :

5 × 1

i)	Linen	a)	Polycamrolactum
ii)	Cotton	b)	Hexamethylene diamin and adipic acid
iii)	Wool	c)	Felting
iv)	Silk	d)	Sericin
v)	Nylon 6, 6	e)	Best fibre
		f)	Rayon
		g)	Kidney shape.

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GROUP – B

(Short Answer Type Questions)

Answer any *three* of the following. $3 \times 5 = 15$

2. Define crimp and contraction. The width of a fabric is 2·5 m, the length of weft yarn after removing from fabric is 2·65 m. Calculate the crimp and contraction % of yarn in weft direction of the fabric.

3. The TM of a 30 Ne cotton yarn is 4·25. Calculate the direct twist factor of the yarn.

4. A loom running at 240 rpm and produce 2/2 twill weave. Calculate the rpm of bottom shaft and tappet shaft.

5. Define the following : Ends, ppi, fabric GSM, temple, drawing.

6. A loom is running with fabric constructional parameter 60 epi and 60 ppi. During the production how can you increase the epi and ppi of the fabric ?

GROUP – C

(Long Answer Type Questions)

Answer any *three* of the following. $3 \times 15 = 45$

7. a) Give a detailed classification of textile fibre. 9

   b) Draw the various cross-sectional shapes of textile fibres which are used in apparel industry. Give at least one example of each. 6

8. a) What do you understand by mono-filament and multi-filament ? Describe the various important primary properties of textile fibre. 2 + 6

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b) What is retting of jute ? Write different field applications of jute fibre. Name three leading jute growing states in India. 2 + 3 + 2

9. a) Justify the following statements : 3 × 3

(i) Wool is preferable as winter garment where linen as summer garment.

(ii) It is very difficult to use jute in wearable clothing.

(iii) Crimped staple fibre yarn has more aesthetical value than filament yarn.

b) Describe the various types of fancy yarns in brief. 6

10. a) Draw the process flow chart of 40s Ne combed cotton yarn. 6

b) Define the various types of fabric. What are the various types of primary, secondary motion in loom ? Describe each in brief. 4 + 5

11. a) Draw the graphical representation of the following weave :

2/1 twill, 2/2 basket, 5 end sateen. 7

b) What are drafting plan and lifting plan ? Draw the design, drafting plan and lifting plan of 3/2 twill. What is the total number of heald frame required to produce this design ? 2 + 5 + 1

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